and along with generations of computer the language were also developed, concurrently.

First Generation Language

The name of the first generation language is called as **'Machine language'**.

What is meant by machine language? Here, the word machine denotes the computer. Basically, this machine is an electrical machine. So, the 'Machine Language' can also be called as 'Electrical language'. How about this Electrical language?

To understand this, let we take a SWITCH as shown in Figure 9.7. In how many modes we can operate a switch?

Normally a switch can be operated in two states.

1. ON – state
2. OFF - state

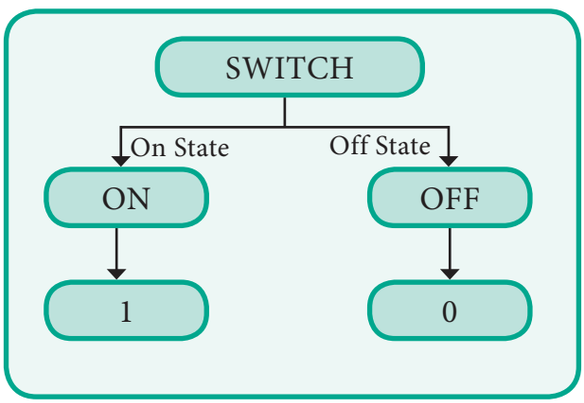


Figure 9.7 **Switch**

NOTE: Switch cannot be operated in any other modes.

When there is supply (ON-state), it is denoted by the letter (numeral) '1' and if there is no supply (OFF-state), it is denoted by the letter '0'.

So, in electrical language, there are only two (1, 0) digits, which are the direct electrical notations and thus machine language is nothing but 1s and 0s. Hence, during the period of first generation computer, they were able to interact with the computer only by using 1s and 0s. During the first generation computer, this machine language was used as shown in Figure 9.8.

Decimal No	Machine Lang
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101

Figure 9.8 **Machine Language**

Since the computer doesn't know anything other than this 1s and 0s, any other alphabets or letters should be converted into this 1s and 0s. This becomes much laborious and tough for the people to understand and follow. Hence, along with second generation computer the second generation language was also developed.

Second Generation Language

The name of the second generation language is **'Assembly Language'** as shown in Figure 9.9. In this, we can use both alphabets and also numbers but only in the form of abbreviations.

Instead of

ADDITION → ADD

SUBTRACTION → SUB

MULTIPLICATION → MUL

DIVISION→DIV.....etc.,

Figure 9.9 **Assembly Language**

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